Lab 9.1 Camera Calibration

In this lab you will try calibrating a camera in two different ways and compare the results.

You will need a smartphone camera for this experiment. If you don't have access to one (or even if you do), it is okay to find someone in the class who does and work as a group.

Follow the instructions below and prepare and submit a document answering the questions in this lab.

1. Look at some images or the live feed from your camera. Without making any measurements, what would you guess is the field-of-view of your camera (along the larger dimension)?
2. You can use the method of similar triangles to estimate the focal length of your camera (and from the focal length, the field-of-view). If you know a) the size of an object, b) the distance to the object, and c) the apparent size of the object in the image, then you can calculate the focal length.

- Write down the formula for calculating the focal length in this manner. Then make measurements of some object (for example, a piece of paper, or a pencil) to estimate the focal length. The easiest way to do this is to frame the object so that it takes up the entire image, e.g., the top of the pencil is at the top edge of the image and the bottom of the pencil is at the bottom edge of the image. Then use the height of the image in pixels as the "apparent size of the object" in your formula, and the distance from the camera to the pencil as the "distance to the object."
- Convert the focal length to field-of-view using the formula presented in class.

3. Take images of a checkerboard calibration pattern with your camera. You don't need to print it out -- it works well to show the checkerboard on your laptop screen. Hold the phone at different angles and positions and take at least 12 images, making sure the entire checkerboard is visible in each image.

- Use the provided notebook to run OpenCV's camera calibration functionality and estimate the intrinsics parameters of your camera. (You can install OpenCV locally with `pip install opencv-python` or run the notebook in Google Colab which has OpenCV pre-installed.)
- Calculate the field-of-view according to OpenCV's estimate of the focal length.
- Compare the results from the checkerboard calibration with your initial guess in step 1 and your manual procedure in step 2.